One Cache, Two Workloads: Certifying KV Compression Along Pointwise and Aggregative Axes, with a Calibrated Expansion Gate

Draft v0.2 — 2026-07-07 — numbers frozen per ABLATION addendum
Artifacts: `memsub-artifacts:/gate-eval/{eval-grid, ablation-results}.json`

Abstract

"Long context" conflates a serving problem—fitting the tokens—with two capability problems that separate the moment full attention becomes unaffordable: *pointwise* recall of exact, low-redundancy facts, and *aggregative* understanding of distributed, redundant content. We argue these are antagonistic optimization targets under any lossy cache compression, and we introduce a certification protocol that scores the two axes separately and forbids blending them. Applying the protocol to untrained KV eviction on Qwen3-4B over a ∼1B-token, four-domain corpus of 98,355 counterfactually filtered questions, we find the axes split almost completely: next-token KL against the full-context teacher stays near 3×10^{-2} while 0.45–0.55 of facts residing in evicted spans are silently lost (measurement noise floor: 0.000). We then show the pointwise axis can be restored by a *decoupled sufficiency gate*: a lightweight probe, trained for ≈$140 on counterfactual sufficiency labels produced by the frozen base model, that decides per query whether compressed memory suffices or exact re-expansion is required. The gate is natively calibrated (test ECE 0.017–0.023), strictly dominates a query-type-routing baseline at every matched expansion rate, and at its operating point cuts silent loss from 0.45–0.55 to 0.146 (bootstrap 95% CI $[0.136, 0.156]$; statistically indistinguishable from the 0.15 target) while expanding on only 16.8% of queries (oracle: 19.4%). A signal decomposition, however, shows the gate's discrimination (evicted-only AUC 0.622) is carried almost entirely by eviction-mechanics features rather than content embeddings—we therefore characterize the shipped gate as a *calibrated mechanics classifier* and identify content-aware sufficiency as open headroom. Two further findings refine the picture: on multi-hop questions the two-workload weakness appears on the *cost* axis rather than the correctness axis—the gate protects correctness by over-expanding 2.3× oracle—and gate discrimination transfers to held-out domains while calibration does not, making per-domain recalibration part of the definition of a deployable gate.

1 Introduction

Benchmarks, methods papers, and product claims routinely report a single "long-context" score. This paper's starting point is that no such single number can be coherent once the context no longer fits in exact form. A needle-class query—an identifier, a figure, a verbatim clause—concentrates its answer in exact bytes with near-zero redundancy against the surrounding context; any lossy transform destroys it at a rate proportional to the compression ratio, and the optimal mechanism for it is selection, indexing, and exact lookup. An integration-class query—summarization, theme tracking, most document reasoning—lives on distributed, redundant information; it is compression-tolerant and often mildly compression-benefited. The two are not

merely different but antagonistic: compression bought for the aggregative workload is paid for in pointwise fidelity, and exact retention kept for needles is memory that compression was meant to free.

If the antagonism is real, three things follow. Evaluation must report the two axes separately, because a blended score lets a method win the aggregate while being needle-dead. Compressed-memory systems need a *routed two-path architecture*—a lossy tier for aggregation and a lossless raw path for pointwise traffic—rather than a single representation with a quality knob. And the router itself becomes the critical learned component: a calibrated, per-query estimate of whether the compressed representation suffices.

We contribute, in order: **(i)** a two-axis certification protocol for compressed LLM memory, with counterfactually filtered probes such that a correct answer proves the information flowed through the (compressed) cache; **(ii)** a measurement at ∼1B-token scale across four document domains showing the axes split under untrained KV eviction—aggregative fidelity essentially intact, pointwise fidelity collapsing, the gap widening with eviction fraction; **(iii)** a decoupled sufficiency gate—trained entirely outside the frozen base model for ≈$140—that restores the pointwise axis at a fifth of the always-expand cost, with calibration good enough to expose as a probability, together with an ablation showing its signal is predominantly eviction mechanics rather than content; **(iv)** two phenomena we believe are documented here for the first time: the multi-hop *cost overhang*, where a well-calibrated single-hop gate protects multi-hop correctness only by systematically over-expanding, and *calibration non-transfer*, where discrimination generalizes across domains but calibrated probabilities do not.

2 Related work

KV-cache reduction. Per-context selection scores tokens at inference and evicts the rest (H2O [12], SnapKV [6], StreamingLLM [10]); amortized selection learns the scorer (KV-Distill [1]); per-context and amortized synthesis write new compact entries rather than keeping original rows (Still [7]); architecture-level lines bake reduction into the model (MLA [2], native sparse attention [11]). Our work is orthogonal to the choice of compressor: we certify compressors along two axes and supply the decision layer their lossiness requires. Notably, the strongest published synthesis results themselves show the axis split we formalize—near-full aggregative utility with needle-retrieval collapse—which we take as independent evidence for the two-workload framing.

Selective prediction and abstention. The gate is a selective-prediction head [3] over an external memory: it estimates, per query, whether the model's conditioning state supports the answer. It relates to answerability and abstention heads in retrieval-augmented settings, with two differences: the label is *counterfactual* (answers under compressed vs. raw conditioning are compared, rather than correctness being judged in isolation), and the head is trained fully decoupled from the backbone, using the backbone only offline as a teacher. Calibration machinery follows Guo et al. [4] and Platt [9].

Memory-augmented agents. OS-style memory hierarchies (MemGPT-class [8]) page text in and out of context but do not certify fidelity or route by workload; our protocol and gate are directly applicable to such systems. Blended long-context suites such as RULER [5] are exactly the evaluations our protocol unbundles.

3 Two axes and a certification protocol

Definitions. Fix a base model f, a compression policy C, and a corpus of (context, question, answer) items where each question is generated from a specific source span. An item is *counterfactually valid* if f answers it correctly with the context and incorrectly without it—so a correct answer under compression proves the information traversed the compressed cache. The **aggregation axis** is measured over integrative behavior: next-token KL between f conditioned on the full cache and f conditioned on C(cache), plus answer-accuracy deltas on items whose source spans survive compression. The **pointwise axis** is the *lost-fact incident rate*: among counterfactually valid items whose source span was compressed away, the fraction answered incorrectly *without* any expansion mechanism—the silent-failure rate.

Protocol. A configuration is certified per axis, never as a blend; the noise floor is established by running the identical probe battery at zero compression; and a configuration certified only on the aggregation axis may ship only behind an active, calibrated expansion gate for pointwise traffic. The last clause converts the antagonism from an observation into a deployment rule. Every report object embeds an evaluation-protocol record: the exact item partition entering each axis, the contents of the compacted context during probing (surviving KV only; no summary text), confirmation that expansion was disabled, and the scoring rule (argmax over answer-letter logits, greedy—format failures are structurally impossible).

4 The axes split: measurement at 1B-token scale

Setup. Base model Qwen3-4B-Instruct-2507 (bf16, greedy, pinned runtime fingerprint). Corpus: 40,000 long documents across financial filings (238M tokens), literature (416M), case law (119M), and source code (154M); 98,355 extractive multiple-choice items generated by the base model from random ∼1k-token sub-chunks, teacher-verified answerable with context and discarded if answerable without (98.4% of the 100k target; code has the lowest filter survival). Equivalence is answer-letter exact match; a manual audit of the complete pointwise-failure population (14 unique items) found 0% format noise, ≈80% genuine forgetting, and 2–3 flawed items. Compression policy: evict-oldest-non-sink at fractions 0.25 and 0.40, with the first four positions pinned as attention sinks [10]. Contexts are 4096-token per-item prefills with item placement randomized by hash, giving natural exposure to eviction.

Result. The zero-eviction control gives a lost-fact noise floor of exactly 0.000 (conditioned on prior-correct probes; the raw metric carries a 0.167 never-knew floor, which is why all headline rates below condition on prior-correct). Under eviction the aggregation axis barely moves—mean KL 0.0303 (f=0.25) and 0.0339 (f=0.40) against the full-context teacher, answer deltas on surviving-span items ≈0—while the pointwise axis collapses: **0.453** (f=0.25, n=1,606) and **0.551** (f=0.40, n=3,493) of facts in evicted spans are silently lost. A matched-random control (same token count evicted uniformly rather than contiguously) inverts the damage profile—4× less pointwise loss at 3× the aggregation KL—showing that recency-structured policies concentrate pointwise damage exactly where aggregate metrics cannot see it. One artifact, two workloads, opposite outcomes.

(Protocol note: an earlier certification pass under packed multi-item contexts with needle-anchored eviction gave 0.708/0.808 *raw; the canonical numbers above are re-stated under the*

Table 1: Signal decomposition (3 seeds, mean ± SD). Features-only matches the full model on the evicted-only task; embeddings add essentially nothing.

Variant	evicted-only AUC	evicted-only PR-AUC	pooled AUC
full model	0.624 ± 0.004	0.622	0.831
features-only (encoders zeroed)	0.621 ± 0.006	0.620	0.837
embeddings-only (features zeroed)	0.536 ± 0.004	0.561	0.557

per-item randomized-placement protocol used throughout this paper, with the reconciliation in Appendix B.)

5 A decoupled sufficiency gate

5.1 Labels, model, calibration

For each of the 98,355 items and each eviction fraction we produce a counterfactual sufficiency label: $y = 1$ iff the answer under the compacted cache differs from the answer under the raw span (196,710 rows; items whose source span survived eviction enter as natural y=0 rows). Splits are grouped by source document, domain-stratified, with an additional leave-one-domain-out rotation. The gate is a dual encoder over frozen Qwen3-Embedding-0.6B—the same embedding model that produces the memory blocks' retrieval shadows—with a 256-d normalized projection interaction, a small feature head (eviction fraction, query–block overlap, evicted flag, domain), and an auxiliary query-workload head. A frozen-embedder linear probe sufficed. Per-fraction temperature search returned T=1.0 for both fractions: the probe is natively calibrated, with test ECE 0.023 (f=0.25) and 0.017 (f=0.40). Marginal cost of the entire gate experiment—labels, training, evaluation, figures—was ≈$140.

5.2 What the gate learned

Pooled test AUC is 0.829, but pooling conflates two tasks: separating evicted from survived rows (near-deterministic given the eviction features) and the question that matters—among evicted spans, which lost their fact. Restricting evaluation to evicted-span rows, the full model reaches AUC **0.622** (95% CI $[0.608, 0.636]$; PR-AUC 0.621)—but the features-only ablation (both encoders zeroed) reaches 0.621 ± 0.006, within seed noise of the full model's 0.624 ± 0.004, while embeddings-only falls to 0.536 ± 0.004 (Table 1). The learned signal is predominantly eviction mechanics (fraction, overlap, position), not content semantics; this also explains the near-tie between document-level and item-level splits observed during training (doc-split AUC 0.8283 vs. leaky item-split 0.8259—there is little content signal for item leakage to exploit). We therefore describe the shipped gate as a *calibrated mechanics classifier* and treat content-aware sufficiency as open headroom.

5.3 Operating point and baselines

At the operating threshold τ=0.488 (chosen on the calibration split), the gate reaches test lost-fact incident rate 0.146 (bootstrap 95% CI $[0.136, 0.156]$ over test items; across 3 training seeds 0.126 ± 0.014) at expansion rate 0.168 $[0.163, 0.173]$, against never-expand at 0.453–0.551 (per

Table 2: Matched-expansion-rate comparison (test split, lost-fact incident rate conditioned on prior-correct).

Expansion rate	0.1	0.2	0.3	0.4	0.5
gate	0.277	0.092	0.000	0.000	0.000
query-type routing	0.470	0.416	0.368	0.308	0.248

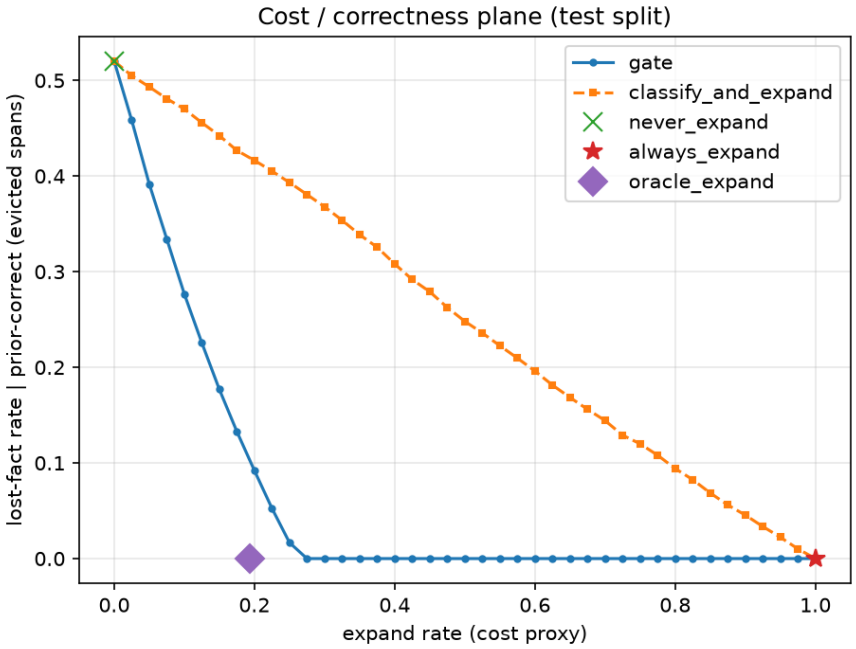


Figure 1: Cost–correctness plane (test split). The gate curve dominates query-type routing everywhere; oracle at $(0.194, 0)$; never-expand at $(0, 0.52)$ pooled.

fraction), always-expand at rate 1.0, and the oracle at 0.000 loss / 0.194 rate. The upper confidence bound marginally exceeds the pre-registered 0.15 target, so we report the frontier rather than claim the threshold: the gate sits at 0.13–0.15 silent loss at ≈0.17 expansion. Against the interpretable baseline—classify the query as pointwise vs. aggregative and expand iff pointwise—the gate strictly dominates at every matched expansion rate from 0.1 to 0.5 (Table 2); the full cost–correctness plane is Figure 1, where the gate's curve lies inside the routing baseline everywhere and reaches ≈0 loss at ∼0.30 expansion rate.

5.4 Multi-hop: the weakness moves to the cost axis

On a 500-item slice of teacher-filtered two-fact questions composed within single documents, the gate at the single-hop threshold *holds correctness*—lost-fact incident rate 0.036 vs. 0.376 never-expand—but does so by over-expanding: recall 0.904 at precision 0.388, expansion rate 0.438 against an oracle of 0.188 (2.3×). The predicted mid-spectrum weakness of retrieval-then-integration queries is real, but it lands on cost, not correctness: a calibrated single-hop gate defends multi-hop questions by firing too often. This cleanly scopes the next component—a selector that scores block *sets* so a multi-hop query triggers one grouped expansion—and gives

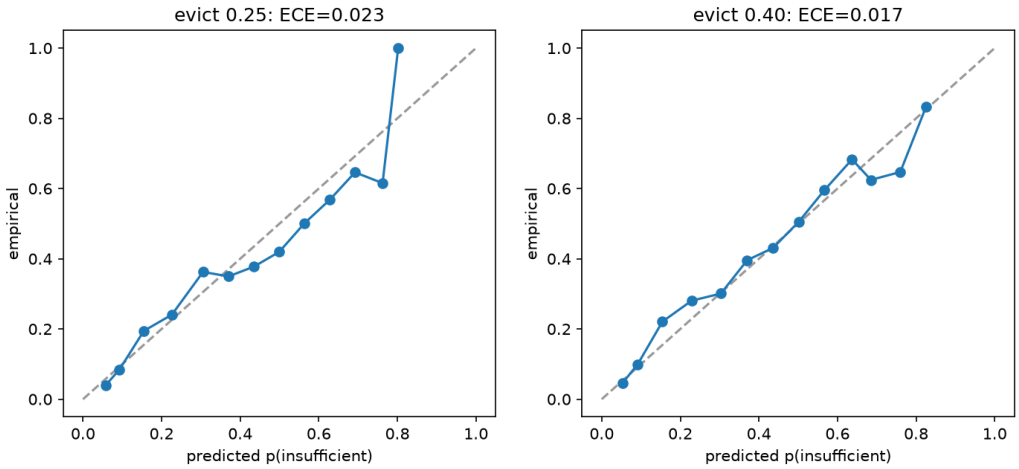


Figure 2: Reliability curves per eviction fraction; the linear probe is natively calibrated (T=1.0; ECE 0.023/0.017).

Table 3: Leave-one-domain-out recalibration on ∼200 items from the held-out domain (frozen gate scores).

Held-out domain	method	ECE before	ECE after
financial	Platt	0.130	0.014
legal	Platt	0.106	0.029
code	temperature	0.064	0.035
literature	Platt	0.116	0.023

it a quantitative target (recover ∼2.3× → ∼1× oracle cost at held recall).

5.5 Calibration does not transfer; discrimination does

Under leave-one-domain-out evaluation the gate's discrimination holds (pooled AUC 0.781–0.865; near-zero lost-fact incident rate at ≈0.26 expansion), but calibration degrades to ECE 0.05–0.13. Fitting on ∼200 labeled items from the new domain repairs it—with a caveat the ablation surfaced: a scalar temperature alone left three of four domains above the 0.05 bar (0.071–0.073); the cross-domain miscalibration has a location component, and escalating to Platt scaling (temperature + bias on frozen scores) restores ECE to 0.014–0.035 in every rotation (Table 3). The practical statement: a "calibrated gate" is not a checkpoint but a checkpoint plus a per-domain recalibration procedure, and deployment to a new domain carries a small, explicit labeling cost.

6 Limitations

Single base model and scale (4B); one compression regime (untrained eviction—trained synthesis compressors will shift the aggregation axis, but published results indicate they do not rescue the pointwise axis, which is the case our protocol and gate address); extractive MCQ probes with a letter-match equivalence function rather than open-ended generation; greedy decoding; per-item

single-turn contexts rather than multi-turn threads; multi-hop limited to k=2 within-document composition. The dead content pathway of §5.2 may be a gloss-quality artifact (extractive chunk prefixes and question-only query encodings) rather than a fundamental limit; probing that requires summary-grade glosses and is left open. The two-axis protocol itself is compressor- and model-agnostic; the specific numbers are not.

7 Conclusion

Single-number long-context evaluation is incoherent under compression: the pointwise and aggregative axes are antagonistic, and at 1B-token scale they split almost completely. The resolution is architectural—a lossy tier for aggregation, a lossless path for needles, and a router between them—and the router turns out to be cheap: a ≈$140 decoupled probe over frozen scores, natively calibrated, dominating query-type routing, and recovering the pointwise axis at a fifth of always-expand cost. The ablation adds the honest mechanism: today that router earns its keep from eviction mechanics, not content understanding, which bounds what it can know and marks content-aware sufficiency as the open problem. What else does not come for free is exactly what the measurements localize: multi-hop queries expose a cost overhang that calls for set-level selection, and calibration is a per-domain property requiring a recalibration procedure. All three are precise targets rather than open mysteries, which is where we think a memory system's decision layer ought to leave its next iteration.

References

[1] Vivek Chari, Guanghui Qin, and Benjamin Van Durme. Kv-distill: Nearly lossless learnable context compression for llms. *arXiv preprint arXiv:2503.10337*, 2025.

[2] DeepSeek-AI. Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024.

[3] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

[5] Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. Ruler: What's the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.

[6] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. Snapkv: Llm knows what you are looking for before generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[7] O'Neill et al. Still: Amortized kv-cache synthesis for long-context inference. *arXiv preprint arXiv:2606.07878*, 2026. Working-notes citation; verify author list and title against the primary.

[8] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[9] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.

[10] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

[11] Jingyang Yuan et al. Native sparse attention: Hardware-aligned and natively trainable sparse attention. *arXiv preprint arXiv:2502.11089*, 2025.

[12] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2o: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

A Evaluation-protocol record

Every certification report object embeds: (i) the item partition per axis—pointwise = needle items whose source chunk lies in the (reference) eviction window; aggregation = a disjoint same-document set, with the count of aggregation items whose chunks fall outside the certification context (they score ≈chance under both conditions and dilute, but cannot bias, the paired delta); (ii) compacted-context contents (surviving KV only; no summary text or auxiliary representation); (iii) expansion disabled during certification probing; (iv) scoring by argmax over answer-letter logits under greedy decoding, making letter-format failures structurally impossible; (v) the KL window length. All arms (control, per-fraction, matched-random) share one probe partition; identical prior-correct counts across arms (21/25) verify it.

B Baseline reconciliation

The earlier certification numbers (0.708/0.808 raw; 0.550/0.650 conditioned) and this paper's canonical never-expand rows (0.453/0.551) measure different protocols. (1) *Context construction*: the earlier pass certified 20 packed contexts whose eviction window was anchored on the needle chunk (adversarial placement; whole chunk always evicted); the canonical protocol uses per-item 4096-token contexts with hash-randomized placement, so evicted rows include boundary items with partial (> 0.5) overlap. (2) *Denominators*: the earlier raw rates include never-knew items (measured floor 0.167); canonical rows have raw-correct ≈1 by construction (generation-time filtering plus greedy determinism), so no never-knew inflation. (3) *Pooling*: the pooled 0.520 mixes both fractions. The paper uses only the canonical per-item protocol; the earlier numbers remain in the certification artifacts flagged as a different protocol.

C Decomposition, seeds, and uncertainty

Table 1 gives the three-variant decomposition (3 seeds each). GT9-class headline metrics across seeds: lost-fact incident rate 0.126 ± 0.014, expansion rate 0.177 ± 0.007. Bootstrap over test items (1,000 resamples): lost-fact incident rate $0.146\,[0.136, 0.156]$, expansion rate $0.168\,[0.163, 0.173]$, evicted-only AUC $0.622\,[0.608, 0.636]$.

D Corpus and filter survival

Four ungated sources (SEC 10-K filings; US court opinions; permissive-license source code; Project Gutenberg literature), 40,000 documents of $\geq \sim$6k tokens, $\sim$927M tokens total. Counterfactual filters retain $\approx$38% of generated items; per-domain yields 25,000/24,790/23,563/25,002 (financial/legal/code/literature). Legal-domain source text carries OCR noise, visible as the weakest per-domain gate recall.

E Reproducibility

Pinned environment (torch 2.7.1, transformers 4.55.2, safetensors 0.5.3); a 14-field compatibility fingerprint (model, tokenizer, chat template, RoPE config, attention implementation, KV layout, quantization, runtime, policy versions, kernel numeric probe) is stored with every KV artifact and validated all-or-nothing. Compute ledger: corpus $\approx$\$500; labels $\approx$\$110; gate training, evaluation, multi-hop slice, and ablations $\approx$\$30. Gate checkpoint, calibration record (including per-domain recalibration entries), evaluation grid, and all per-item predictions are stored as content-addressed artifacts.